CALIFORNIA STATE UNIVERSITY
LONG BEACH

# Lecture 9
# Interaction Concepts

# What is interaction?

# Classes of interaction techniques

- **navigation**—user controls for altering the position of the camera and for scaling the view (what gets mapped to the screen) such as panning, rotating, and zooming.
- **selection**—user controls for identifying an object, a collection of objects, or regions of interest to be the subject of some operation, such as highlighting, deleting, and modifying.
- **filtering**—user controls for reducing the size of the data being mapped to the screen, either by eliminating records, dimensions, or both.
- **reconfiguring**—user controls for changing the way data is mapped to graphical entities or attributes, such as reordering the data or layouts, thereby providing a different way of viewing a data subset.

# Classes of interaction techniques

- **encoding**—user controls for changing the graphical attributes, such as point size or line color, to potentially reveal different features.
- **connecting**—user controls for linking different views or objects to show related items.
- **abstracting/elaborating**—user controls for modifying the level of detail.
- **hybrid**—user controls combining several of the above in one technique, for example, increasing the screen space assigned to one or more focus areas to enable users to see details, while showing the other areas of data in a smaller space, in a way that preserves context.

# Overview

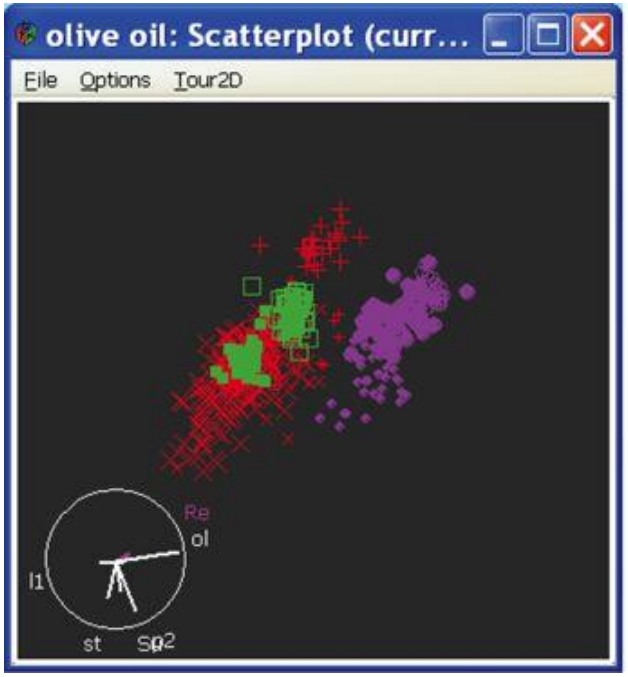
1. Interaction Operators
2. Interaction Operands and Spaces
3. A Unified Framework

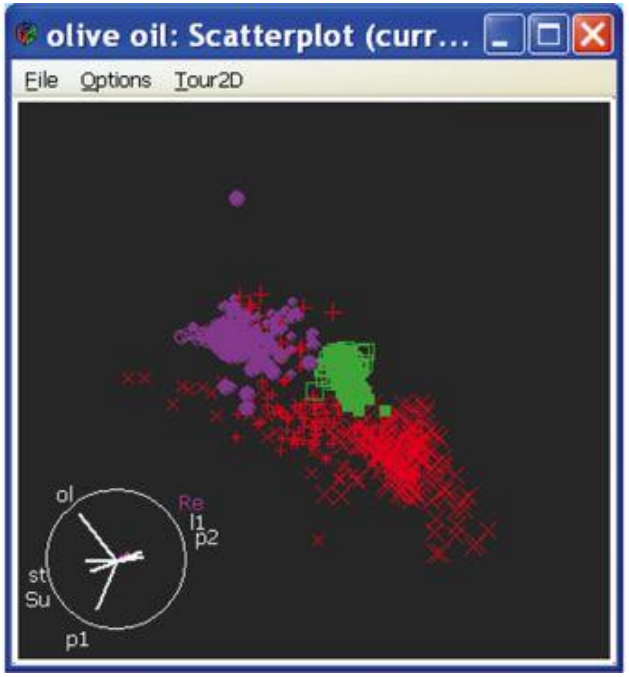
# 1. Interaction Operators

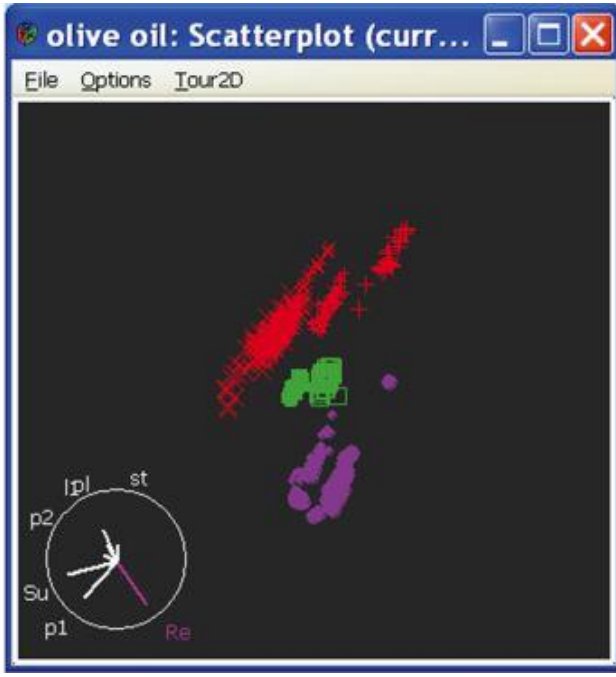
# 1.1. Navigation Operators

- Navigation (also sometimes referred to as exploration) is used to search for a **subset** of data to be viewed, the **orientation** of this view, and the **level of detail** (LOD).
- The subset in question may be one that is recognized by some visual pattern or one on which further or more detailed exploration is desired.
  - E.g. In a typical three-dimensional space, this can be specified using a camera location, a viewing direction, the shape and size of the viewing frustum, and an LOD indicator.
  - E.g. In multiresolution visualizations, LOD changes can correspond to drilling down or rolling up hierarchical representations of the data.
- Navigation operators can work in absolute or relative coordinates within their particular spaces. Incremental navigation may have different granularities, depending on whether the user wants a small or significant change.

# An Example




Views of the grand tour for the grades data set using GGobi.

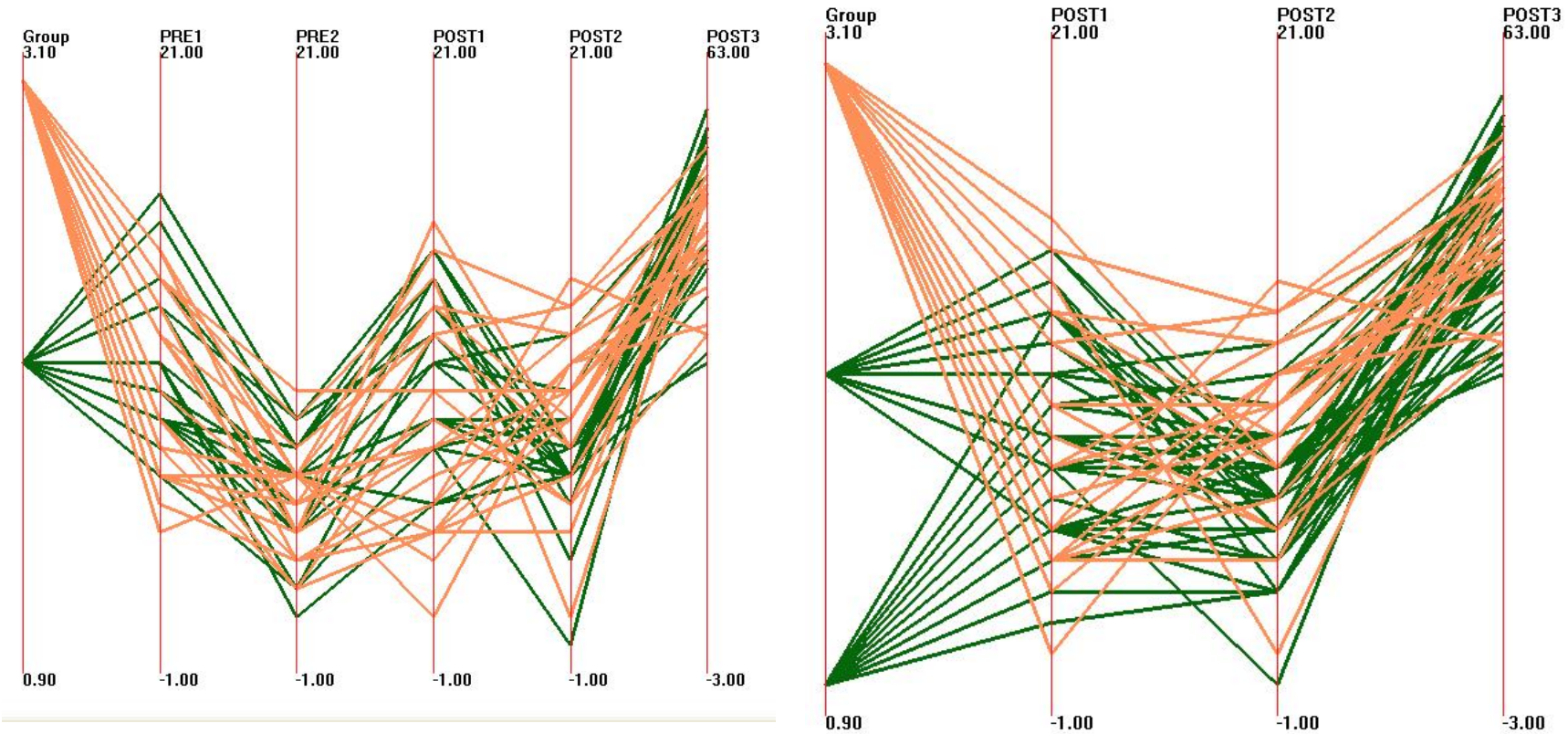
# 1.2. Selection Operators

- In selection, the user **isolates** a subset of the display components, which will then be subjected to some other operation, such as highlighting, deleting, masking, or moving to the center of focus.
- Selections may be performed in an indirect manner, where the system selects elements that match a user's input set of constraints.

# 1.3. Filtering Operators

- Filtering, as the name implies, **reduces** the volume of data to be visualized by setting constraints specifying the data to be preserved or removed.
- The distinction between filtering and selection followed by deletion or masking is a subtle, but important point.
  - **Filtering**, in general, is most often done in an ***indirect*** manner, e.g., the filter specification is not performed on the data visualization itself, but via a separate interface or dialog box. In fact, filtering is often done prior to viewing the data, to avoid overloading the data display.
  - **Selection** is most often done in a ***direct*** manner, by indicating objects on the visualization via mouse motions, for example. The operation performed on the selected subset can result in a view that is indistinguishable from a filtering operation.

# An Example




XmdvTool (http://davis.wpi.edu/xmdv/)

Filtering rows and columns of the grades data set using XmdvTool.

# 1.4. Reconfiguring Operators

- Reconfiguring the data within a particular visualization can often be used to expose features or cope with complexity or scale.
- By **reorganizing** the data, such as filtering some dimensions and reordering those that remain, different views are provided to the user.
- For example,
  - a powerful tool with table-based visualizations is to **sort** the rows or columns of the data to highlight trends and correlations.
  - Other types of reconfiguration might be to change the **dimensions** being used to control the x- and y-coordinates of a plotted marker, or even to derive positions based on transformations of the data.
  - Popular instances of this include the use of principal component analysis (PCA) or multidimensional scaling (MDS) to lay out data points so that relationships among all the dimensions are best conveyed in the 2D layout.

# 1.5. Encoding Operators

- ***Recoding*** can provide the user a library of possible different types of visualization; features of the data that are difficult or impossible to see with one such mapping might become quite apparent in another.
- Many visualization tools today support ***multiple types of visualization***, because no single visualization is effective for all tasks that need to be carried out by the user. Each visualization is most suitable for a subset of data types, data set characteristics, and user tasks.
- Other forms of encoding operations include those that modify the color map used, the size of graphical entities, and their shape. These can be considered *variations* within a particular type of visualization, and can be used to emphasize or reveal features of interest.

# Multiple Synchronized Views: an example

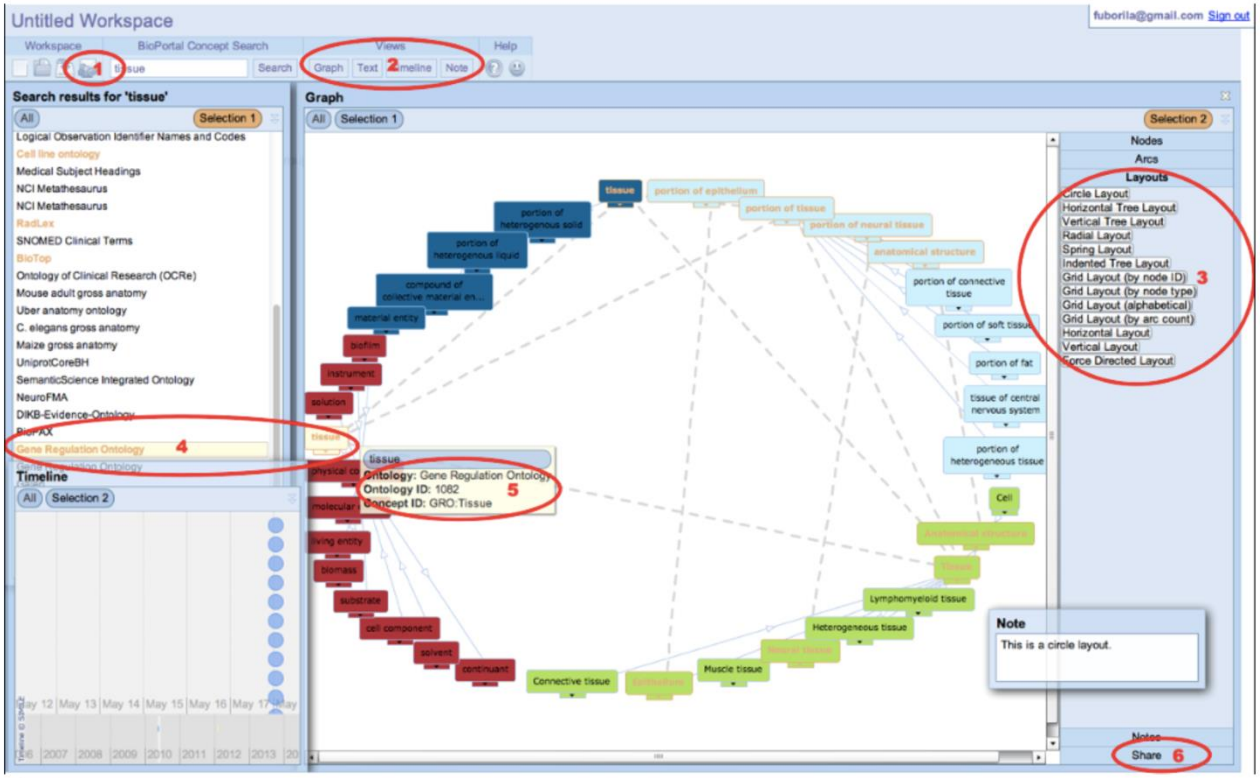
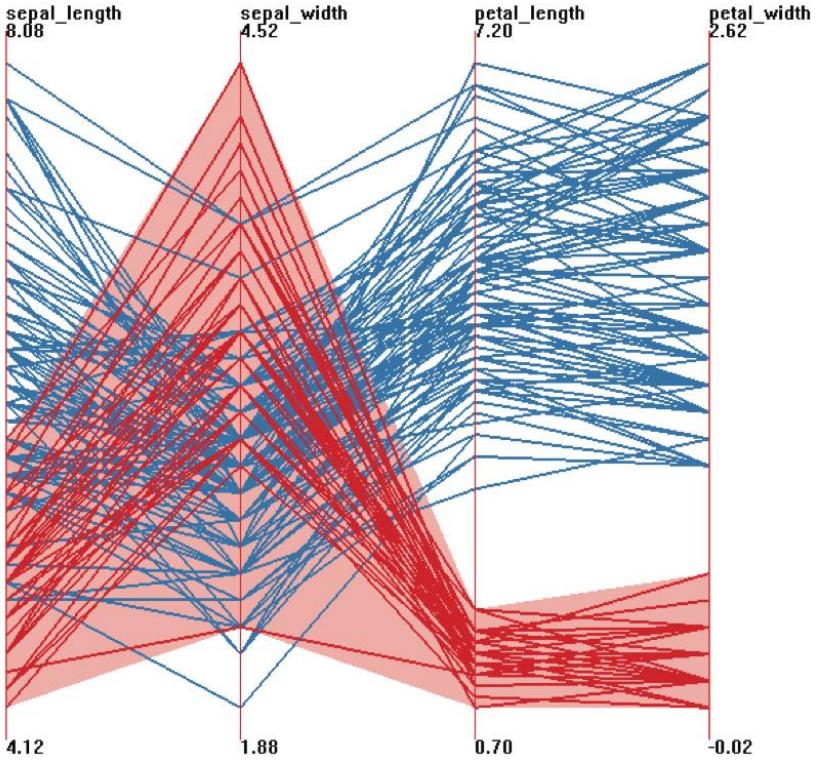


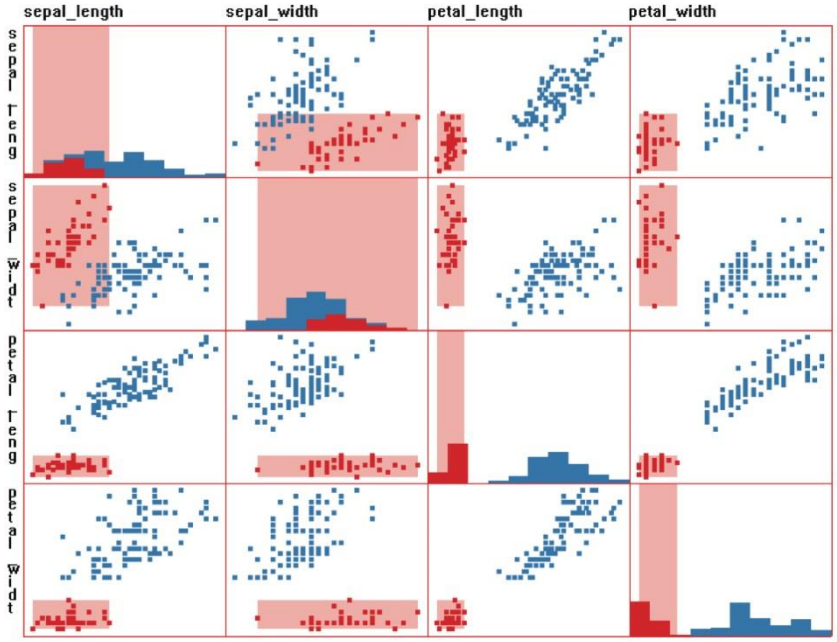

**Fig. 1.** An Example of Collaborative Ontology Visualization in BioMixer. The user is presented with four tabs: *Workspace*, *BioPortal Concept Search*, *Views* and *Help*. There are four buttons under *Workspace*, including (from left to right) *Create New* workspace, *Load* existing workspace, *Save* and *Share* a workspace. The current release contains four views (circle 2), including *Graph*, *Text*, *Timeline* and *Note*. A variety of visualization layouts (circle 3) are supported in the Graph View. The user can send an existing workspace to collaborators by using the *Share* feature (circle 1 & 6) and add comments to visualizations by using the *Note* view (bottom-right). The user can also obtain an iFrame to embed a single visualization in external websites (see Fig. 4). Details of nodes (circle 5) are displayed on demand (i.e. when a user hovers over a node) in BioMixer. In addition, coordination (circle 4) is achieved by highlighting the node/selection under the mouse cursor across multiple views. Note that the user is free to visualize any selection (i.e. any combination of nodes) by dragging and dropping it into the new view.

# 1.6. Connection Operators

- A frequent use for selection operations is to **link** the selected data in **one view to** the corresponding data in **other views**.
- While other forms of connection between subwindows of an application exist, such as when opening a new data file, *linked selection* is probably the most common form of communication between windows found in modern visualization tools.
- When the selection data is allowed to be interactively changed, the operator is called ***brushing***, in which case the user is continuously changing the selection in one view, and the corresponding linked data in one or more other views is highlighted. The resulting interactive and dynamic display provides information about the changes in values in the linked displays.

# An Example




XmdvTool (http://davis.wpi.edu/xmdv/)

Filtering rows and columns of Example of linked brushing. A cluster is isolated in parallel coordinates, with linked selection in the scatterplot matrix. Selected data is dark, while brush extents are shown as light bands or rectangles.

# 1.7. Abstraction/Elaboration Operators

- In dense data and information displays, it is often desirable to focus in on a subset of the data to acquire details (**elaboration**) while reducing the level of detail (**abstraction**) on other parts of the data set.
- One of the most popular techniques of this type is using distortion operators, e.g. panning and zooming. The distortion may take place within the original visualization, or may appear in a separate window. Distortions vary in the features that are preserved and the amount of context maintained.

# Exercise: Integrating Interactive Features

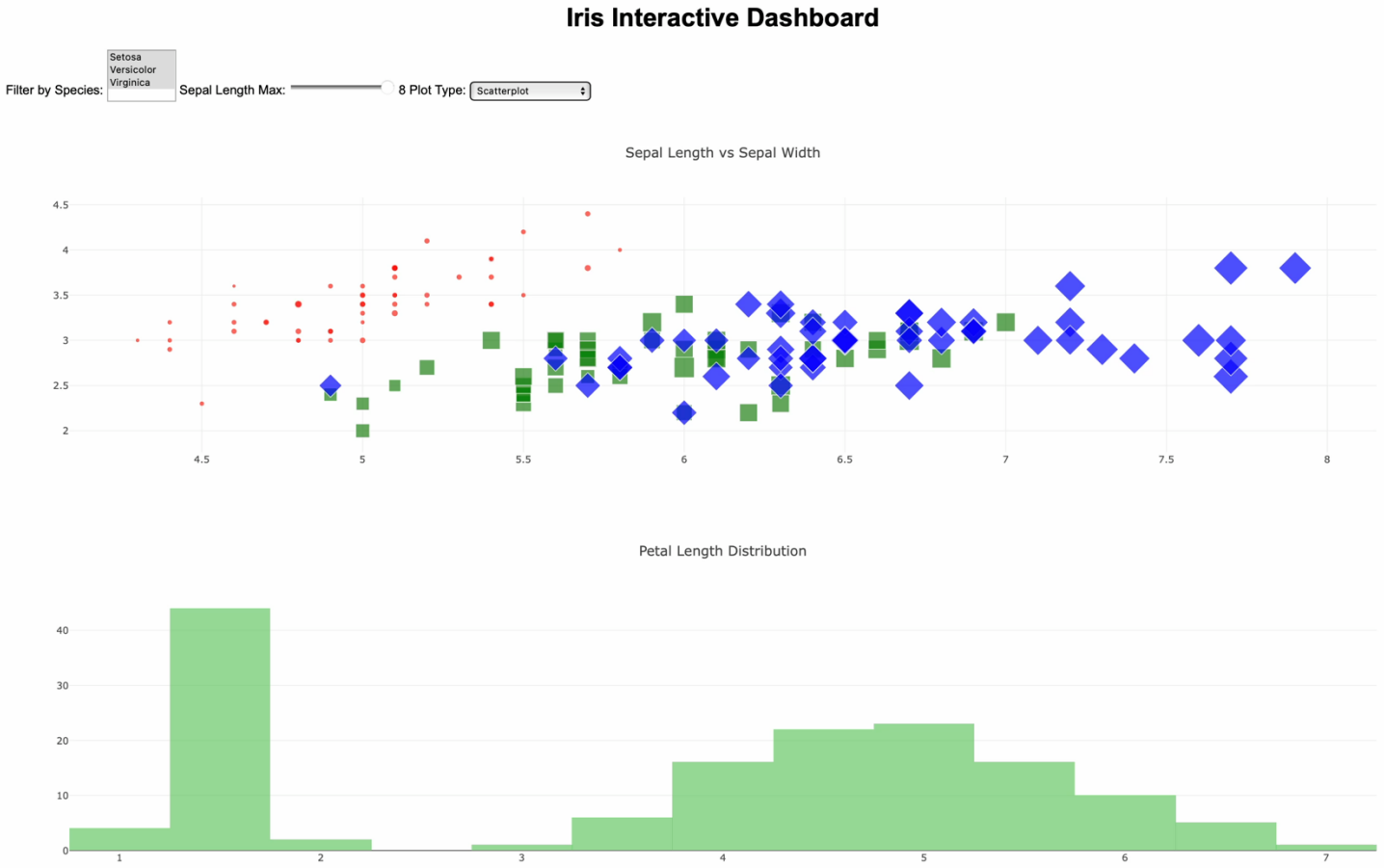
Using the **Iris dataset** to generate a scatterplot with interactive features, such as:

- Navigation: Implement panning, zooming, or rotating on a scatterplot (e.g., allow users to zoom into clusters of points, or rotate a 3D plot）.
- Selection: Enable click or lasso selection of points; highlight selected points and display their details in a side panel.
- Filtering: Add dropdown menus or sliders to filter data by category (e.g., species in Iris); numeric range (e.g., sepal length).
- Reconfiguring: Allow users to switch between layouts, e.g., Scatterplot vs. parallel coordinates; reorder axes dynamically.
- Encoding: Provide controls to change color schemes, point sizes, or marker shapes based on different attributes.
- Connecting: Implement linked views, selecting points in one plot highlights them in another (e.g., scatterplot and histogram).
- Abstracting/Elaborating: Add a detail-on-demand feature, hovering over a point shows detailed info; clicking expands a panel with full record details.
- Hybrid Technique: Implement a focus+context view, e.g., Zoom into a selected region while keeping the rest of the data visible in a smaller overview.

Suggested **Tools**: plotly.js

# Exercise: an example

- Starter code on Canvas



# 2. Interaction Operands and Spaces

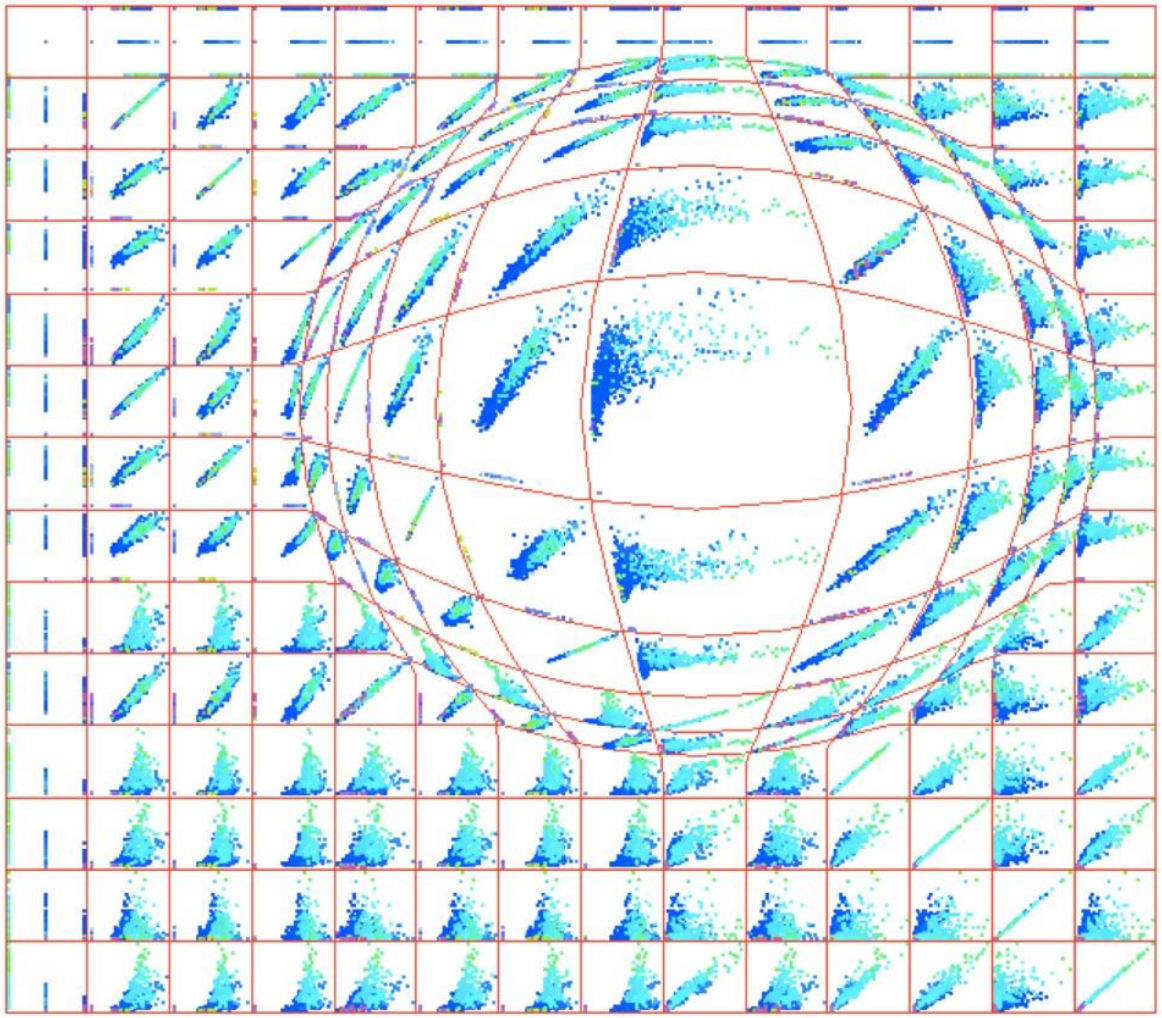
# What's an interaction operand?

- An interaction operand is the section of space upon which an interactive operator is applied.
- To determine the result of an interactive operation, one needs to know within what space the interaction is to take place.
  - In other words, when a user clicks on a location or set of locations on the screen, what entities does he or she wish to indicate? Possibilities include the pixel(s), the data value or record mapped to the location, or even the component of the visualization structure (e.g., an axis) at or near that location.

# 2.1. Screen Space (Pixels)

- Screen space consists of the pixels of the display. Navigation in screen space typically consists of actions such as panning, zooming, and rotation. Note that in each case, no new data is used; the process consists of pixel-level operations such as transformation, sampling, and replication.
- ***Pixel-based selection*** means that at the end of the operation, each pixel will be classified as either selected or unselected. Selection areas may also be contiguous or non-contiguous. Screen space filtering consists of cropping or masking a region of pixels.

# An Example



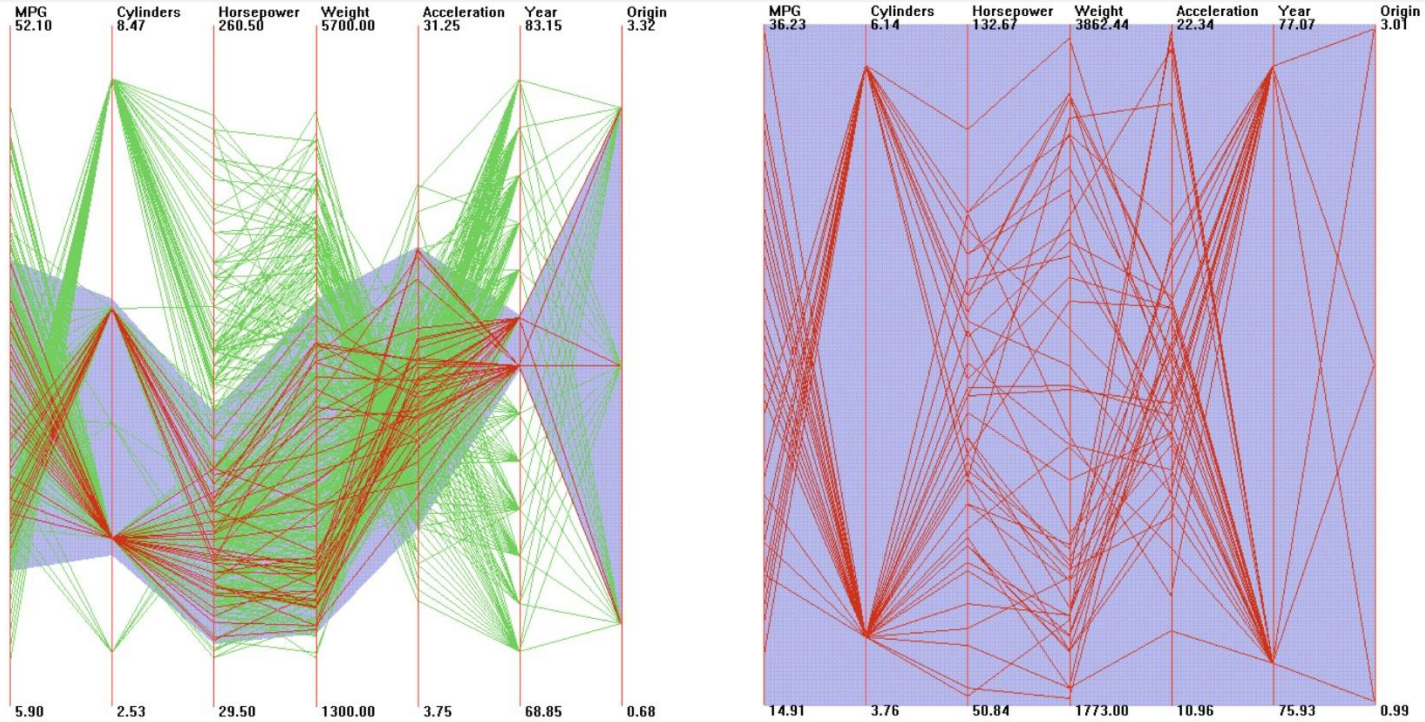
In screen space techniques, pixel regions are enlarged or reduced to provide selective detail. In this scatterplot matrix display, a center of focus has been selected and magnified using a confocal lens technique.

Matthew O. Ward and Jing Yang. "Interaction Spaces in Data and Information Visualization." In Eurographics Symposium on Visualization (Vis-Sym), edited by Oliver Deussen, Charles D. Hansen, Daniel A. Keim, and Dietmar Saupe, pp. 137–145. Aire-la-Ville, Switzerland: Eurographics Association, 2004.

# 2.2. Data Value Space (Multivariate Data Values)

- Operations on data space are applied directly to the data, rather than to the screen.
- Navigating in data value space involves using the data values as a mechanism for view specification.
  - The analogous operations for panning and zooming would be to change the data values being displayed;
  - panning would shift the start of the value range to be shown,
  - while zooming would decrease the size of this range.

# An Example of data value space distortion



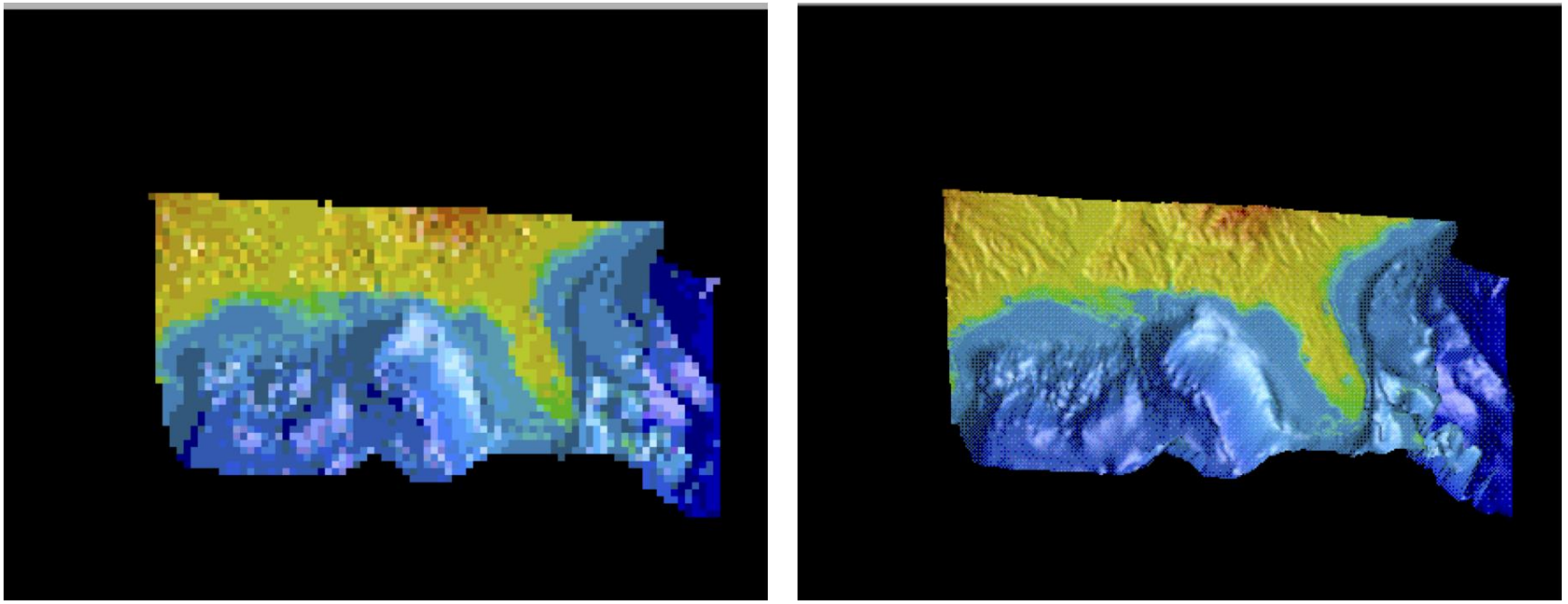

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In data value space distortion, transformations are performed according to the dimensionality of the data. In this example, an *N*-dimensional hyperbox is selected via painting over a section of an axis and scaled in all dimensions (by different amounts) to fill a unit hypercube, which is then displayed. Animation is used to preserve context. Clusters and anomalies within the selected region are much easier to see in the zoomed version.

# 2.3. Data Structure Space (Components of Data Organization)

- Data can be structured in a number of ways, e.g. lists, tables, grids, hierarchies, and graphs.
- For each structure, one can develop interaction mechanisms to indicate what portions of the structure will be manipulated, and how this manipulation will be manifested.
- Navigation in data structure space involves moving the view specification along the structure, as in showing sequential groups of records, or moving down or up a hierarchical structure (as in drill-down and roll-up operations).

# An Example

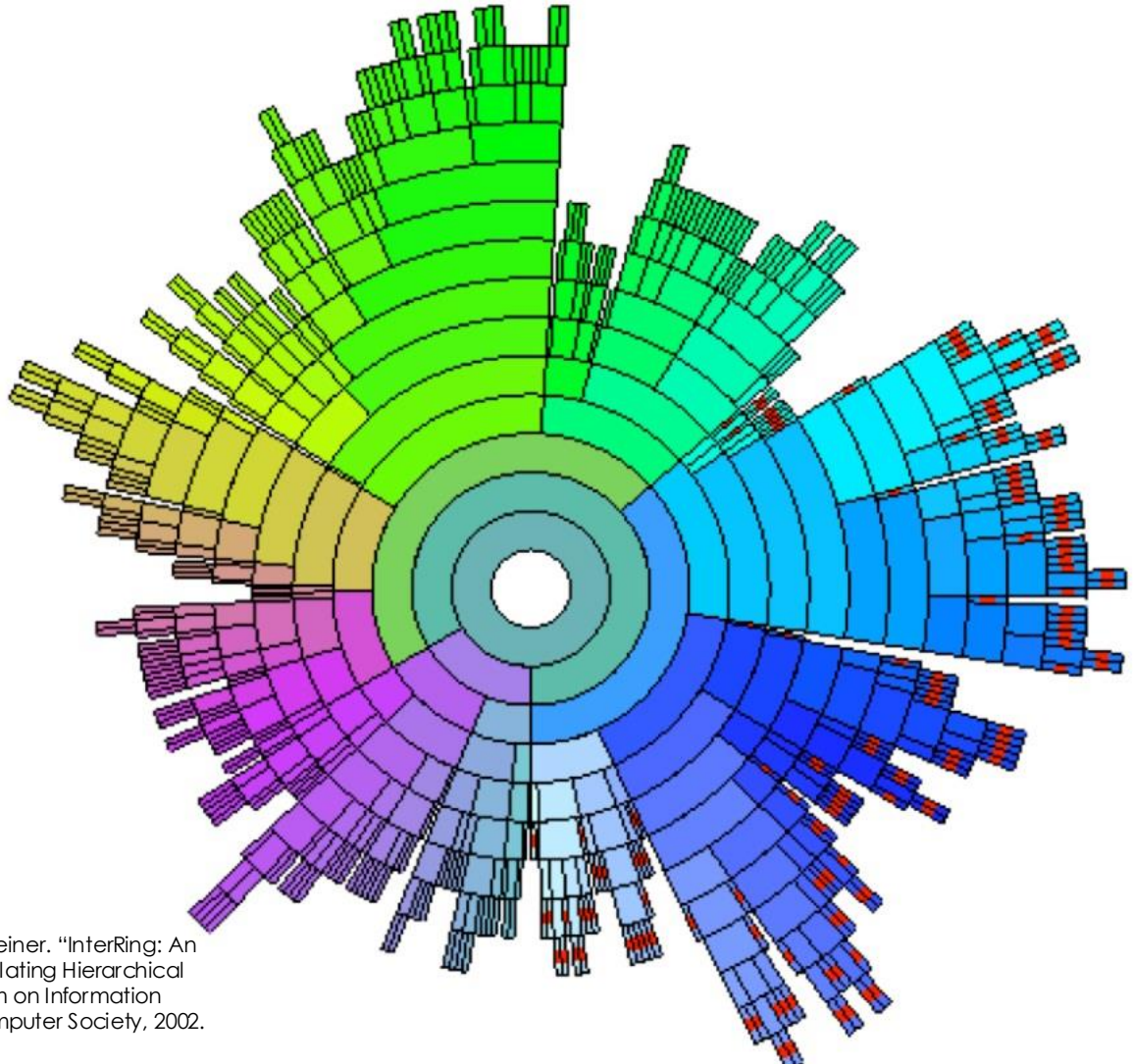


OpenDX (http://www.opendx.org/)

In screen space zooming (left), pixels are replicated to provide selective size, while in data space zooming (right), the data itself can be resampled at the appropriate resolution.

- Selection in data structure space generally involves displaying the structure and allowing the user to identify regions of interest within it.
- This in turn can drive the display of the data corresponding to the selected substructure. For example,
  - structure-based brushing involves controlling the selection of data stored in a cluster hierarchy, with interactions such as highlighting data that fall within a particular branch of the tree.
  - InterRing (next slide) is a radial space-filling hierarchy visualization tool that allows semi-automatic selection of nodes, according to the hierarchical structure.

Selection of nodes in a hierarchy via InterRing. Nodes with a red stripe in them have been selected via a user-specified query, rather than one node at a time.

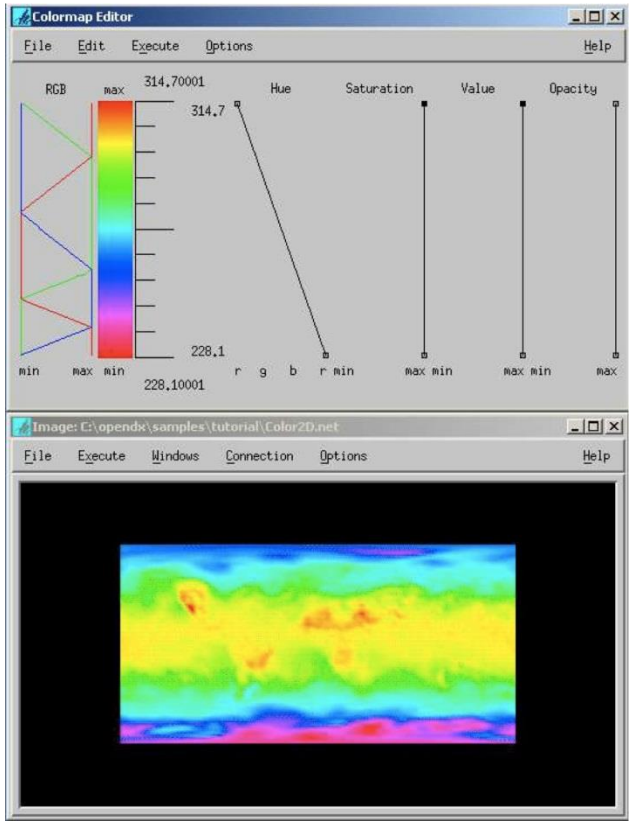


Jing Yang, Matthew O. Ward, and Elke A. Rundensteiner. "InterRing: An Interactive Tool for Visually Navigating and Manipulating Hierarchical Structures." In Proceedings of the IEEE Symposium on Information Visualization'02, pp. 77–84. Washington, DC: IEEE Computer Society, 2002.

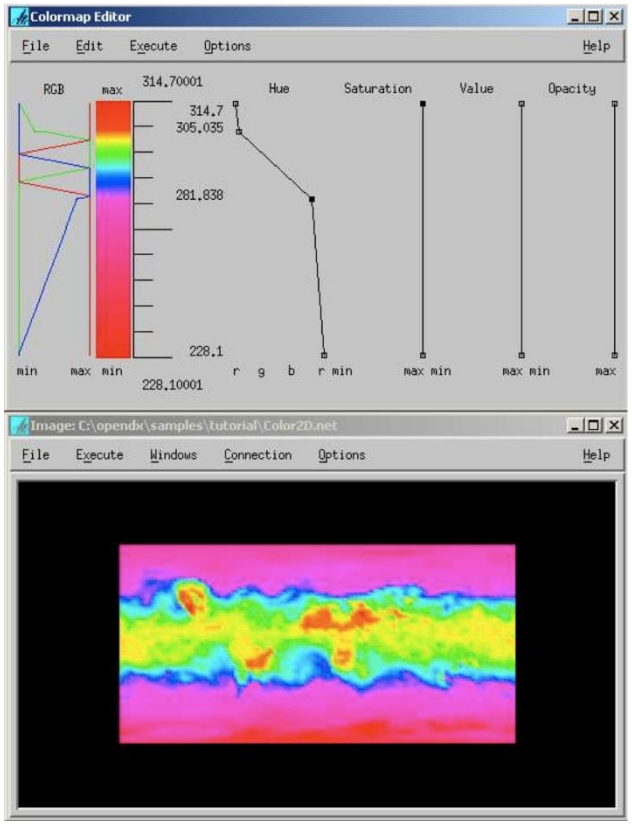
# 2.4. Attribute Space (Components of Graphical Entities)

- In attribute space, operators are focused on one or more of the attributes associated with the graphical entity being used to convey information. Such attributes could include color, size, shape, or any other of the eight visual variables.
- Navigation in attribute space is similar to that in data value space (i.e. panning involves shifting the range of the values of interest, while zooming can be accomplished by either scaling the attributes or enlarging the range of values of interest).
- Attribute-space selection requires the user to indicate the subrange of a given attribute of interest. For example,
  - given a visual depiction of a color map, a user can select one or more entries to highlight.
  - if data records have attributes such as quality or uncertainty, a visual representation of these attributes, accompanied by suitable interaction techniques, can allow users to filter or emphasize data according to the attributes.
  - Remapping is often done in attribute space, either via selecting different ranges of an attribute to be used in the data to graphic mapping, or by choosing a different attribute to be controlled by the data.

# An Example




OpenDX (http://www.opendx.org/)




OpenDX (http://www.opendx.org/)

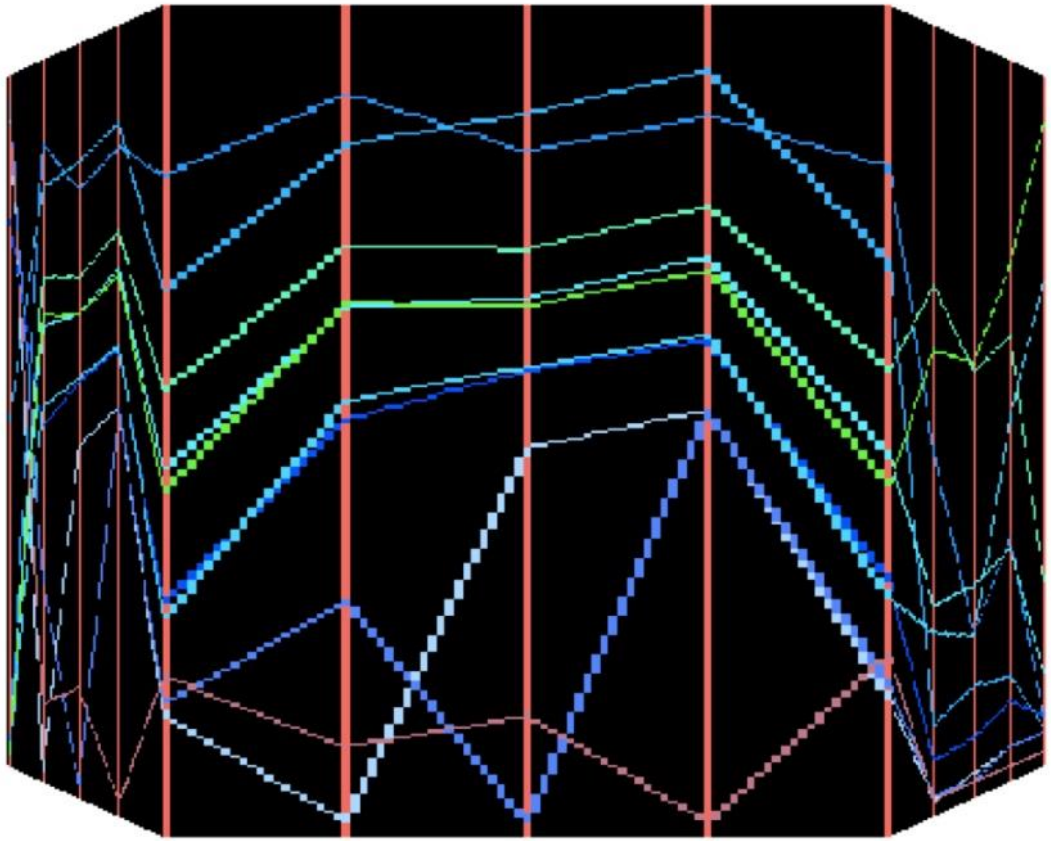
Attribute-based distortion modifies one or more attributes of the graphical objects used to depict the data, as shown in this colormap modification. The color map is distorted to allot a greater portion to values in the middle of the data range.

# 2.5. Object Space (3D Surfaces)

- In object space displays, the data is mapped to a geometric object, and this object (or its projection) can undergo interactions and transformations.
- Navigation in object space often consists of moving around objects and observing the surfaces on which the data are mapped.
- The system should support global views of the object space as well as close-up views. The latter may be constrained to enable the user to find good views more quickly. Selection involves clicking anywhere on the object(s) of interest, or indicating target objects from a list.

# An Example

Object-based techniques distort an object upon which data has been projected. In this example, inspired by the perspective wall, a parallel coordinates display is projected onto walls, and perspective is used to make a selected wall more readable, while maintaining context with the rest of the data.



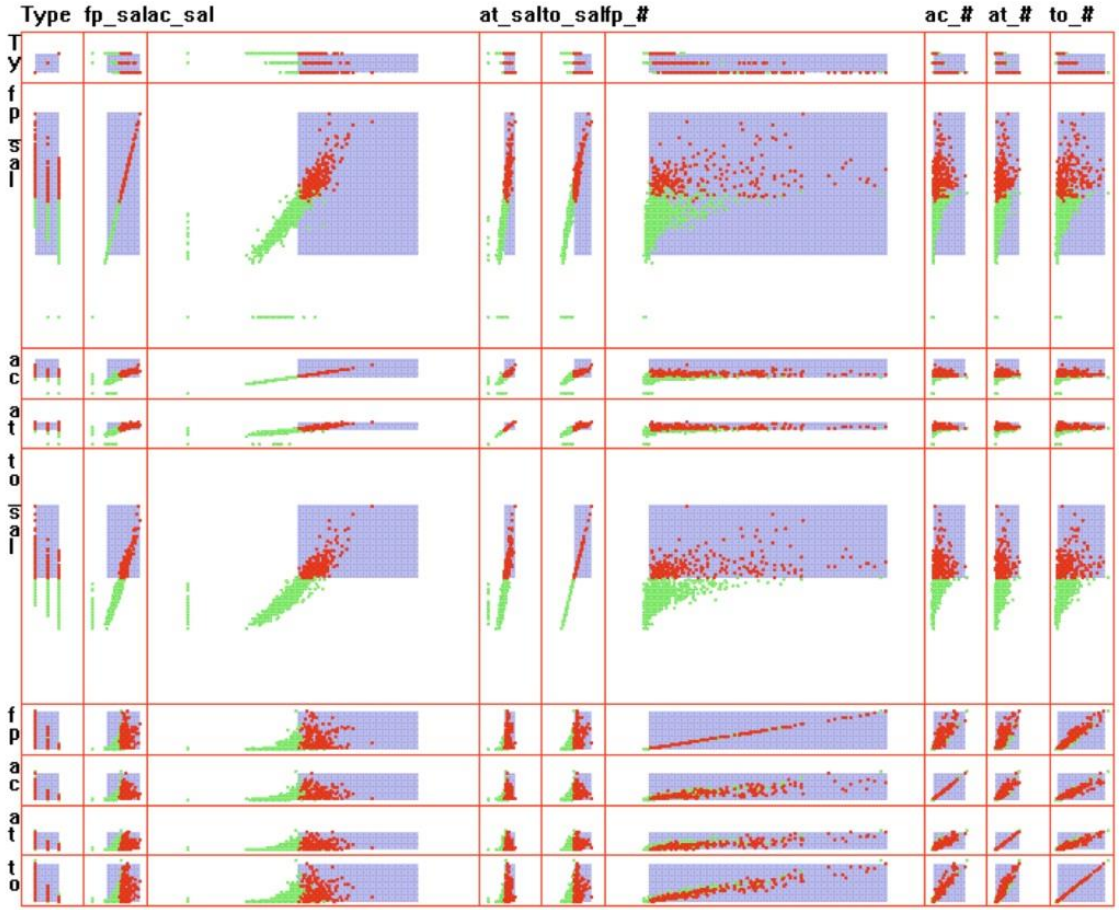
Matthew O. Ward and Jing Yang. "Interaction Spaces in Data and Information Visualization." In Eurographics Symposium on Visualization (Vis-Sym), edited by Oliver Deussen, Charles D. Hansen, Daniel A. Keim, and Dietmar Saupe, pp. 137–145. Aire-la-Ville, Switzerland: Eurographics Association, 2004.

# 2.6. Visualization Structure Space

- A visualization consists of a structure that is relatively independent of the values, attributes, and structure of data.
- For example, the grid within which a scatterplot matrix is drawn and the axes displayed in many types of visualizations are each components of the visualization structure, and can be the focus of interactions.
- Visualization *dashboards*, consisting of a set of tightly or loosely connected visualizations juxtaposed on the screen, have become increasingly popular for monitoring and analyzing complex data environments, such as for business analytics and command-and-control centers.

# An Example

Structure-based distortion modifies the underlying structural elements of the visualization. This example shows a scatterplot matrix with two grid cells (and their corresponding rows and columns) magnified, with a corresponding shrinkage in other cells.




Matthew O. Ward and Jing Yang. "Interaction Spaces in Data and Information Visualization." In Eurographics Symposium on Visualization (Vis-Sym), edited by Oliver Deussen, Charles D. Hansen, Daniel A. Keim, and Dietmar Saupe, pp. 137–145. Aire-la-Ville, Switzerland: Eurographics Association, 2004.